

Managing information - means working out what information is needed by the people with whom they work, where it might come from and what they need it for. It involves perceiving information as a resource which your organization has, and which can be consciously used and reused to meet its need

Information management is a cycle of processes that support the organization's learning activities: identifying information needs, acquiring information, organizing and storing information, developing information products and services, distributing information, and using information

According to a process view of information management, IM is a continuous cycle of six closely related activities:

- identification of information needs;
- acquisition and creation of information;
- analysis and interpretation of information;
- organization and storage of information;
- information access and dissemination;
- information use

Collection of project information can take many forms, such as written, video, oral, audio or electronic.

Storage is important for a range of purposes including: analysis, identifying historical trends, developing lessons learned, satisfying legislative requirements etc.

Curation includes the process of gathering and organizing information relevant so that it can add value.

Dissemination involves consideration of questions such as: what information is to be distributed, to whom, in what format, how often, under what circumstances and using what security protocols?

Archiving takes place after a period of time, usually determined by a mix of company policy and judgement

Destruction eventually takes place because the information is no longer deemed important, especially in comparison to the on-going storage costs and legislation e.g. privacy laws, only allows the storage of information for a prescribed period
Information management encompasses systems such as:

- document management (DM)
- records management (RM) - web content management (CM)
- digital asset management (DAM)
- learning management systems (LM)
- intranet platforms
- collaboration
- enterprise search

Ten principles:

- Principle 1: recognize (and manage) complexity
- Principle 2: focus on adoption
- Principle 3: deliver tangible & visible benefits
- Principle 4: prioritize according to business needs
- Principle 5: take a journey of a thousand steps
- Principle 6: provide strong leadership
- Principle 7: mitigate risks
- Principle 8: communicate extensively
- Principle 9: strive for a seamless digital employee experience (DEX)
- Principle 10: choose the first project very carefully

Data - unorganized raw facts that need processing without which it is seemingly random and useless to humans
an individual unit that contains raw material which does not carry any specific meaning.
doesn't depend on information.
measured in bits and bytes
never suited to the specific needs of a designer.

Information - processed, organized data presented in a given context and is useful to humans.

a group of data that collectively carry a logical meaning.
depends on data.

measured in meaningful units like time, quantity, etc.
specific to the expectations and requirements

Database management system (DBMS) is a collection of programs that manages the database structure and controls access to the data stored in the database.
Database is a shared, integrated computer structure that stores a collection of **End-user data**, that is, raw facts of interest to the end user and **Metadata**, or data about data, through which the end-user data are integrated and managed.
Databases can be classified according to the number of users, the database location(s), the expected type and extent of use, and structure.

Types of DBMS

A single-user database supports only one user at a time.

A multiuser database supports multiple users at the same time

Centralized database supports data located at a single site

Distributed database supports data distributed across several different sites

Operational database designed primarily to support a company's day-to-day operations
Data warehouse focuses primarily on storing data used to generate information required to make tactical or strategic decisions.

Unstructured data are data that exist in their original (raw) state, that is, in the format in which they were collected.

Structured data are the result of taking unstructured data and formatting (structuring) such data to facilitate storage, use, and the generation of information.

Database design refers to the activities that focus on the design of the database structure that will be used to store and manage end-user data.
Decomposition - designing appropriate data repositories of integrated information using the two-dimensional table structures found in most databases
Manual File Systems - often manual, paper-and-pencil systems.
Computerized File Systems - a data processing (DP) specialist was hired to create a computer-based system that would track data and produce required reports.
File System Redux: Modern End-User Productivity Tools - Personal computer spreadsheet programs such as Microsoft Excel are widely used by business users, and allow the user to enter data in a series of rows and columns so that the data can be manipulated using a wide range of functions

Data Dictionary Management - DBMS stores definitions of the data elements and their relationships (metadata) in a data dictionary.
Data Storage Management - One of the DBMS functionalities is creating and managing the complex structures required for data storage, thus relieving the difficult task of defining and programming the physical data characteristics.
Data transformation and presentation - The DBMS transforms entered data in to required data structures.
Security Management - The DBMS creates a security system that enforces user security and data privacy.
Multi User Access Control - uses sophisticated algorithms to ensure that multiple users can access the database concurrently without compromising the integrity of the database.
Backup and Recovery Management - perform routine and special backup and restore procedures.
Data Integrity Management - The DBMS promotes and enforces integrity rules, thus minimizing data redundancy and maximizing data consistency.

Data redundancy exists when the same data are stored unnecessarily at different places.
Data inconsistency exists when different and conflicting versions of the same data appear in different places.
Data integrity is defined as the condition in which all of the data in the database are consistent with the real-world events and conditions.
Data anomalies define anomaly as “an abnormality”
Update anomalies
Insertion anomalies
Deletion anomalies

Structured Query Language (SQL) is the defacto query language and data access standard supported by the majority of DBMS vendors.

Data modeling - refers to the process of creating a specific data model for a determined problem domain.

Problem domain - a clearly defined area within the real-world environment, with well-defined scope and boundaries that is to be systematically addressed.

Data model is a relatively simple representation, usually graphical, of more complex real-world data structures.

Model is an abstraction of a more complex real-world object or event.

Entity - anything about which data are to be collected and stored

Attribute - a characteristic of an entity

Relationship - describes an association among entities

One-to-many (1:M) relationship

Many-to-many (M: N or M:M) relationship

One-to-one (1:1) relationship

Constraint - a restriction placed on the data.

Business rule - a brief, precise, and unambiguous description of a policy, procedure, or principle within a specific organization.

Hierarchical Model - Basic logical structure is represented by an upside-down “tree” It contains Levels or Segments.

Network Model - Created to represent complex data relationships more effectively than the hierarchical model, to improve database performance and to impose database standard.

Schema - which is the conceptual organization of the entire database as viewed by the database administrator

Subschema - which defines the portion of the database “seen” by the application programs that actually produce the desired information from the data contained within the database.

Data management language (DML), which defines the environment in which data can be managed and to work with the data in the database

Data definition language (DDL), which enables the database administrator to define the schema components.

Relational Model - In this relation (sometimes called table) is composed of intersecting rows and columns.

Each **row** is called a **tuple**.

Each **column** represents an **attribute**

Relational Diagram- Representation of entities, Attributes & relationships.

Relational table - stores collection of related entities.

Entity Relationship Model - Graphical representation of entities and their relationships in a database structure

Entity relationship diagram (ERD) - Uses graphic representations to model database components

Object-Oriented (OO) Model - Data and relationships are contained in a single structure known as an object

Object - Contains operations such as changing its data values, finding a specific data value, and printing data values

Class is a collection of similar objects with shared structure (attributes) and behavior (methods).

Methods - a set of procedures

Invoicing problem - invoices are generated by customers, each invoice references one or more lines, and each line represents an item purchased by a customer

Object/Relational and XML - Primarily geared to business applications
Emerging Data Models: Big Data and NoSQL - **Big Data**: Find new and better ways to manage large amounts of Web- generated data and derive business insight from it, **NoSQL databases**: Not based on the relational model, hence the name NoSQL
External Model - End users' view of the data environment
Conceptual Model - Represents global view of the entire database
Internal Model - Representation of the database as “seen” by the DBMS
Physical Model - Operates at lowest level of abstraction

Logical database view - how the data appear to the user to be stored.

Relation - logical view of the relational database is facilitated by the creation of data relationships based on a logical construct

Tables are the basic building blocks of a relational database.

Keys – defined by functional dependencies

Superkey - An attribute (or combination of attributes) that uniquely identifies each row in a table.

Candidate key - A minimal (irreducible) superkey. A superkey that does not contain a subset of attributes that is itself a superkey

Primary key - A candidate key selected to uniquely identify all other attribute values in any given row. Cannot contain null entries.

Secondary key - An attribute (or combination of attributes) used strictly for data retrieval purposes.

Foreign key - An attribute (or combination of attributes) in one table whose values must either match the primary key in another table or be null

Relational schema is a textual representation of the database tables where each table is listed by its name followed by the list of its attributes in parentheses.

Integrity Rules are imperative to a good database design.

Entity Integrity - All primary key entries are unique, and no part of a primary key maybe null.

Reference Integrity - A foreign key may have either a null entry, as long as it is not a part of its table's primary key, or an entry that matches the primary key value in a table to which it is related.

SELECT - also known as **RESTRICT**, yields values for all rows found in a table that satisfy a given condition.

PROJECT yields all values for selected attributes.

UNION combines all rows from two tables, excluding duplicate rows.

INTERSECT yields only the rows that appear in both tables.

DIFFERENCE yields all rows in one table that are not found in the other table; that is, it subtracts one table from the other.

PRODUCT yields all possible pairs of rows from two tables—also known as the Cartesian product.

JOIN allows information to be combined from two or more tables.

Equijoin, links tables on the basis of an equality condition that compares specified columns of each table.

Inner join is a join that only returns matched records from the tables that are being joined.

Outer join, the matched pairs would be retained, and any unmatched values in the other table would be left null.

Data dictionary provides a detailed description of all tables found within the user/designer-created database.